

Input and Output

27 April 2026 11:05

Sometimes a program needs to interact with the users to get some input data or information from the end user and process it to give the desired output. In Python we have the input () function for taking the user input. The input () function prompts the user to enter data. It accepts all user input as string. The user may enter a number of string but the input () function treats them as string only. The syntax for input() is:

```
input([prompt])
```

Prompt is the string that we may like to display on the screen prior to taking the input and it is optional. When a prompt is specified first it is displayed on the screen, after which the user can enter the data. The input () Takes exactly what is typed from the keyboard, converts it into a string and assign it to the variable on the left hand side of the assignment operator (=). Entering data for the input function is terminated by pressing the Enter key.

```
fname = input('Enter your first name: ')
Enter your first name: Shiv
age = input('Enter your age: ')
Enter your age: 17
type(age)
<class 'str'>
```

```
age = int(input('Enter your age: '))
Enter your age: 18
type(age)
<class 'int'>
Here 'age' is in int data type.
```

Python uses the **print()** function to output data to standard output device - the screen. The function print () evaluates the expression before displaying it on the screen. The print () outputs a complete line and then moves to the next line for a subsequent output. The syntax for print () is:

```
print(value[,..., sep = ' ', end = '\n'])
```

- **sep:** The optional parameter **sep** is a separator between the output values. We can use the character integer or a string as a separator. The default separator is space.

For example:

```
fname
'Shiv'
age
18
st
'abcdecbac dba kcsb'
```

```
print(fname, age, st)
Shiv 18 abcdecbac dba kcsb
```

← sep default is space

```

print(fname, age, st)
Shiv 18 abcdecba dba kcsb
print(fname, age, st, sep='$')
Shiv$18$abcdecba dba kcsb
print(fname, age, st, sep='gamma')
Shivgamma18gammaabcdecba dba kcsb
print(fname, age, st, sep='#')
Shiv#18#abcdecba dba kcsb

```

← sep default is space

← separator between two content.

- **end:** This is also optional and it allows us to specify any string to be appended after the last value. The default is a new line. For example:

```

fname = 'Shiv'
lname = 'Ray'
print(fname, end=' ')
print(lname)

```

Type Conversion

```

a = 18
b = 22
a + b
40
r = a + b

```

```

r
40
type(r)
<class 'int'>

```

```

r = a / 4
r
4.5

```

← implicit type conversion done by Python.

```

type(r)
<class 'float'>

```

```

c = 'ten'
r = a + c

```

← Python is unable to do conversion.

Traceback (most recent call last):

```

File "<pyshell#46>", line 1, in <module>
    r = a + c

```

TypeError: unsupported operand type(s) for +: 'int' and 'str'

```

r = c + a

```

← Python is unable to do conversion.

Traceback (most recent call last):

```

File "<pyshell#47>", line 1, in <module>
    r = c + a

```

TypeError: can only concatenate str (not "int") to str

Explicit conversion: When the conversion is done by the programmer's consent then it is called explicit conversion. For example:

```

c
'ten'
a
18
type(c)
<class 'str'>
type(a)
<class 'int'>
r = c + str(a)
r
'ten18'

```

explicit conversion is done here.

↑ int

It down, explicit conversion, also called type casting, happens when data type conversion takes place because the programmer forced it in the program. The general form of an explicit data type conversion is:

```
(new_data_type)(expression)
```

With explicit type conversion, there is a risk of loss of information since we are forcing an expression to be of a specific type. For example, converting a floating value of X equal to 20.67 into an integer type that is. In the (X) Will discard the fractional part 0.67. Following are some of the function in Python that are used for explicitly converting an expression or a variable to a different type.

Function	Description
int(x)	Converts x to an integer.
float(x)	Convert x to a floating point number.
str(x)	Convert x to a string representation.
boolean(x)	Convert x to a boolean representation.
complex(x)	Convert x to a complex representation.

Homework:

1. Basic Input & Output

Write a program to input your **name**, **age**, and **city** and display them in the following format:

Name: ____

Age: ____

City: ____

2. Type Identification

Write a program that takes two inputs:

- one integer
- one string

Print both values and display their **data types** using **type()**.

3. String to Integer Conversion

Take a number as input using **input()** and:

- Convert it into an integer
- Print its square

4. Separator Usage

Write a program to input three values: **name**, **marks**, **grade** and print them using:

- default separator
- # as separator
- *** as separator

5. End Parameter

Write a program to print:

Hello World

Welcome to Python

on the **same line** using the end parameter.

6. Arithmetic with Input

Take two numbers as input from the user and display:

- Sum
- Difference
- Product
- Division

Ensure proper type conversion.

7. Implicit Type Conversion

Write a program:

- Assign integer value to a variable
 - Divide it by another number
 - Print the result and its type
- Explain why the type changed.

8. Error Handling Concept

Write a program where:

- a = 10
- b = "5"

Try:

```
print(a + b)
```

Then fix the error using proper type conversion.

9. String Concatenation with Numbers

Take a number as input and concatenate it with the string "Number entered is: "

Example output:

Number entered is: 25

10. Mixed Conversion Problem

Write a program that:

- Takes **length** and **breadth** as input (as strings)
- Converts them into integers
- Calculates and prints the **area** and **perimeter**